Geographical Dual-Prior Guided Few-Shot Network for Cross-Domain Hyperspectral Image Classification

Weihuan Deng, Huiting Li, Qiqi Zhu, *Senior Member, IEEE*, and Qingfeng Guan

***Abstract*— Cross-domain hyperspectral image (HSI) classification (HSIC) addresses the challenge of real-time labeling of new regions. To mitigate the performance decline caused by unseen classes, a few-shot learning (FSL) method is used. However, these methods fail to fully consider the problem of sample scarcity and classification imbalance due to FSL methods. In addition, the issue of category confusion stemming from localized spectral fluctuations within the same class is commonly overlooked. To solve these problems, a geographical dual-prior guided few-shot network (Gprior-FSN) is proposed. In Gprior-FSN, combining prior knowledge of the first law of geography, a geographical prior guided bicorrelated (G-B) sample enhancement mechanism is proposed which includes geospatially correlated enhancement (GCE) and spectral feature correlated enhancement (SFCE). GCE uses a hierarchical sampling strategy to tackle the inherent imbalance problem for FSL methods. Subsequently, GCE mitigates sample scarcity via neighborhood sample expansion while identifying candidate pseudosamples with geospatial correlation. To make the acquired pseudosamples of the same category bicorrelated in both geospatial and spectral features, G-B combining spectral feature clustering and probabilistic statistics mechanism is designed. Inspired by the second law of geography, Gprior-FSN uses a spatial constraint mechanism to effectively enhance intraclass similarity by reducing spatial local heterogeneity, while improving global interclass discriminability. Finally, to further capture representative spatial–spectral feature, a weighted dual feature fusion network is designed. Experimental results from three distinct HSI datasets show that Gprior-FSN outperforms advanced HSIC methods in both efficiency and accuracy. In addition, the Gprior-FSN demonstrates strong generalization performance on real GF-5 image.**

***Index Terms*— Classification imbalance, cross domain, few-shot learning (FSL), geographical, local spectral fluctuation, sample scarcity.**

Received 4 June 2024; revised 12 September 2024; accepted 15 October 2024. Date of publication 11 November 2024; date of current version 23 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42271413; and in part by the Key Laboratory of Natural Resources Monitoring in Tropical and Subtropical Area of South China, Ministry of Natural Resources under Grant 2024NRMK02. *(Corresponding author: Qiqi Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3495525

I. Introduction

HYPERSPECTRAL image (HSI) is extensively adopted for environmental surveillance [1], [2], target positioning [3], [4], military surveillance [3], [5], and topographic classification [6]. HSI classification (HSIC) involves assigning predefined category labels to individual pixels, making it a crucial task in HSI applications [7]. In the early period, HSIC methods mainly focused on spectral feature extraction approaches, simply applying classifiers to pixels vector, such as support vector machines (SVMs) [8], neural network [9], and logistic regression [10]. However, the presence of spectrum uncertainty increases the probability of misclassification [11]. Numerous feature extraction methods that incorporate spatial and spectral features are already established, including morphology [12], [13], filtering [14], [15], and coding based methods [16]. Great results can be obtained through these traditional HSIC methods, but they rely extensively on manually designed or shallow-based descriptors [17].

Due to the powerful feature extraction capabilities of deep learning, deep-learning-based HSIC tasks are extensively used in recent years [18], [19]. Although deep learning has achieved a great success in HSIC, adequate labeled samples are normally required [20], [21]. However, obtaining the HSI samples consumes a lot of manpower and material resources [22], [23]. Some unsupervised and semi-supervised learning methods are used to overcome labeled data-limited problems, which effectively use both labeled and unlabeled data to enhance classifier performance [24]. Fang et al. [25] applied a sampling rank-order clustering method to address the issue of sample scarcity by assigning pseudolabels to unlabeled samples. Pu et al. [26] use octave convolution to capture local–global information of unlabeled samples. Although above unsupervised and semi-supervised methods can mitigate sample scarcity problem, these methods are mainly performed on the same data distribution [27].

In real-time fine mapping applications, it is extremely difficult to perform large-scale real-time labeling for a new region [28], [29]. Cross-domain classification methods provide potential solutions for the challenge of labeling new regions [30]. However, in scenarios where target domain (TD) samples are extremely limited, models tend to perform well in capturing source domain (SD) features but struggle with TD features. Moreover, due to the obvious differences between the two domains, a direct transformation method

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often leads to performance degradation [31]. Adversarial-based methods can extract domain-invariant features and have been widespread used in cross-domain tasks involving HSIC. Wang et al. [32] proposed local spatial congruence and spectral alignment to facilitate knowledge transfer. Zhang et al. [33] introduced a network for expanding domains in cross-scene HSIC. Although these adversarial-based methods achieved good classification results, domain adaptation methods often suffer from significant performance degradation when new classes emerge in the TD [34].

In practical applications, especially facing fine HSIC task for new areas, it is difficult for the existing data to cover all ground object categories. Few-shot learning (FSL) methods, which can effectively solve this problem using very few training samples in the TD, have attracted widespread attention [35]. Dong et al. [36] introduced a novel FSL-based approach to the problem of class differences between SD and TD. Wang et al. [37] introduced FSL to solve unseen classes problem and proposed classwise attention. By looking at the classification results in Fig. 1(a), we counted the classification accuracies of ten FSL methods on Indian Pine (IP), Pavia University (UP), and Salinas and found that the tail class is significantly better than the head class. It can note that although the FSL method performs well in solving the unseen class problem, using the same number of training samples for each class, it results in the head category receiving a smaller number of training samples. This makes it difficult for the model to adequately capture the feature variations in the head classes, which leads to the optimization direction of the model being biased toward the tail classes. Specifically, as shown in IP dataset, the classification accuracies of head category 1 and head category 7 shown in the green box in Fig. 1(a) are obviously lower than those of tail category 2 and tail category 11 shown in the red box. The head category refers to the category with a larger coverage area, while the tail category refers to the one with a smaller coverage area. The head classes are more abundant in the image, but the relatively small number of available labels makes it difficult for the model to adequately capture the feature changes in these categories during training. The problem of sample scarcity of TD and classification imbalance caused by FSL methods has always existed but has not received sufficient attention. Therefore, designing networks to solve this problem is important to promote the application of FSL methods in cross-domain HSIC tasks.

HSIs have an obvious advantage in fine classification tasks due to their rich spectral information, which can obtain a continuous spectral curve of the ground objects. However, motivated by the local spatial heterogeneity of the second law of geography, spatial intervals create differences between classes. As shown in Fig. 1(b), spectral curve is often affected by localized spatial fluctuations heterogeneity within classes, leading to intraclass inconsistencies and inadequate interclass discrimination [38]. By reducing local spectral fluctuations in TD, the model preserves the diversity and integrity of SD features, avoiding fitting bias due to excessive smoothing in SD. This approach enhances intraclass consistency and interclass discrimination within TD, facilitating more effective knowledge transfer. Studying the low-rank properties of HSI

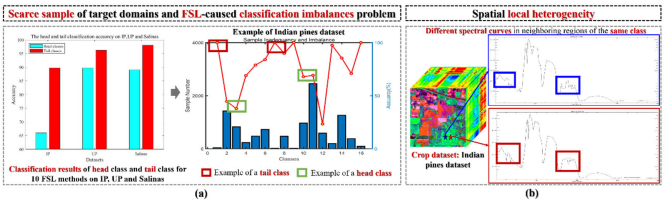


Fig. 1. Two key problems. (a) Problem of sample scarcity and FSL-caused classification imbalance problem. By analyzing the classification results of the ten FSL methods, the results show that the classification performance of the tail classes is significantly better than that of the head classes. (b) Issue of spatial local heterogeneity. The inconsistency of spectral curves in neighboring regions within a class often results in heightened intraclass variation, consequently diminishing interclass discriminability.

will help mitigate the effects of local spectral fluctuations, which in turn will improve the classification performance [39]. Sumarsono and Du [40] proposed a method that segments HSIs into multiple patches and processes each patch individually using a robust low-rank matrix representation model. However, the traditional block-based segmentation methods cannot capture complex local spatial structures with sufficient accuracy [41]. The superpixel segmentation method is capable of acquiring multiple uniformly similar spectral feature regions and achieving spatially adaptive segmentation. Yang et al. [42] presented a superpixel-guided discriminative low-rank representation of HSIs to reduce local spectral volatility. Although the previously mentioned methods achieve satisfactory classification results, local spectral fluctuations often affect the classification in cross-domain hyperspectral classification tasks. Therefore, it is very challenging to capture comprehensive features and mitigate the negative image of local spectral fluctuation in cross-domain HSIC.

To solve the above problem, a geographical dual-prior guided few-shot network (Gprior-FSN) is proposed which includes geographical prior guided bicorrelated (G-B) sample enhancement mechanism and geographical prior inspired weighted dual (G-W) feature fusion network. First, G-B sample enhancement mechanism is designed to get enough high-quality pseudosamples and balance the number of per class. Minimum noise fraction (MNF) is used for reducing the data dimensionality. Then, G-W feature fusion network was introduced to reduce local spectral fluctuations and capture more comprehensive features. Finally, we use the conditional domain discriminator (CDD) to reduce domain shift.

The key contributions in this article are listed below.

1) Gprior-FSN is proposed for cross-domain HSIC. To solve the problem of sample scarcity and FSL-caused classification imbalance, G-B sample enhancement mechanism is designed to get high-quality bicorrelated pseudosamples and increase the number of samples in the head classes. The G-B sample enhancement mechanism reduces the sample dependency and increases model convergence speed. G-W feature fusion network is propped to improve intraclass consistency.
2) To obtain bicorrelated pseudosamples of geospatial and spectral features, G-B sample enhancement mechanism is proposed based on the priori geoscientific knowledge of the first law of geography. The strategies of neighborhood sample enhancement and class hierarchies are used

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for obtaining geospatially correlated candidate pseudolabels. To capture spectral features and geospatially correlation, we conduct superpixel segmentation on the image and assign each superpixel block to the category with the highest frequency.

3) To reduce the impact of spatial local heterogeneity, the local spatial constraints strategy is designed to minimize the effect of local spectral feature fluctuations. To comprehensively perceive the HSI features, a two-branch network of weighted local spatially constrained features fused with the original image features is designed. G-W feature fusion network exhibits a comprehensive understanding of spectral features, effectively enhancing both intraclass similarity and interclass discrimination.

To validate the effectiveness of Gprior-FSN, this article conducts experiments on three datasets containing agricultural and urban scenarios. The experimental results show that our method significantly improves both accuracy and efficiency. For verifying the generalization of Gprior-FSN, we conducted generalization experiments on the real GF5 hyperspectral dataset. The experimental results show that Gprior-FSN has significant classification performance.

The subsequent sections of this article are outlined as follows. Section II provides a description of the related work. Section III explains our framework. Section IV is an introduction to the HSI and experimental analysis. The conclusion of this article is presented in Section V.

II. Related Works

Given that this article is based on domain-adapted, semi-supervised, and FSL HSIC, the current state of the research on domain-adapted, semi-supervised HSIC methods, and FSL HSIC is described below.

A. Domain-Adapted HSIC

Due to substantial disparities in data distributions between the two domains, direct transfer methods often yield poor results. Therefore, mitigating domain shift poses a highly challenging task [43]. Maximum mean discrepancy (MMD) as an effective nonparametric distance metric is widely used in the strategies to measure the discrepancy between the SD and TD distributions [44]. The smaller the MMD value, the higher the similarity between the two domains. Zhang et al. [45] adopted MMD for marginal distribution alignment of cross-domain HISC task. Zhu et al. [46] proposed the concept of subdomains and used local MMD for subdomain alignment. The formula for MMD is as follows:

$$\mathrm{MMD}(X_s, X_t) = \left\| \frac{1}{N_s} \sum_{j=1}^{N_s} \psi\left(x_j^S\right) - \frac{1}{N_T} \sum_{i=1}^{N_T} \psi\left(x_i^T\right) \right\|_H^2. \quad (1)$$

While MMD offers flexibility and a solid theoretical foundation for measuring domain discrepancy, its effectiveness is heavily influenced by kernel selection [47], [48]. Domain adaptation is an effective approach to reduce domain shift across domains which have widely applied in cross-domain HSIC. Ye et al. [49] propose a dictionary-learning-based feature-level domain adaptation technique for cross-scene HSIC. Ma et al. [50] proposed an unsupervised cross-dataset HSIC method with an adversarial domain-based adaptation. Domain adversarial networks reduce domain shift through adversarial training, allowing the use of unlabeled data from the TD, but their complexity and instability can lead to poor performance on unseen categories.

B. FSL for HSIC

The growing numbers of researches have attempted to close the gap between human learning and machine learning, which has given rise to a new subfield of machine learning called FSL [51]. In the classification field, FSL aims to learn an effective classification model applying a small number of labeled training labels [52]. The FSL methods have also been widely used in the field of HSIC. Zhang et al. [53] introduced a prototype network-based FSL for HSIC. FSL methods have been adopted in cross-domain HSIC tasks for their capacity to effectively address the unseen class challenge with a limited set of training samples. Zhang et al. [54] combined FSL and domain adaptation networks. Peng et al. [55] combined transformer and FSL for a cross-domain HSIC task. Ye et al. [56] enhanced cross-domain FSL by adaptively weighting the conditional adversarial loss based on class variation. While progress has been made in addressing unseen classes, the persistent issue of class imbalance in FSL methods, where some classes are underrepresented, remains a significant challenge.

C. Semi-Supervised HSIC

Semi-supervised learning is a prominent research focus in HSIC. The classifier performance is effectively enhanced through the utilization of unlabeled data. Fang et al. [25] applied a sampling rank-order clustering method to address the issue of sample scarcity by assigning pseudolabels to a large number of unlabeled samples. Guan et al. [57] used hyperpixel clustering to extract spectral feature-associated pseudolabels and subsequently introduced a pseudolabel correction module to precisely align pixel-level data with hyperpixel-clustered pseudolabels. Fang et al. [25] introduced a deep clustering method to group deep spectral features and created pseudolabels for a large number of unlabeled samples. Ma et al. [58] presented multidecision labeling to decide a sample label. Tan et al. [24] combined spatial neighborhood information with a classifier to increased pseudosamples. Wang et al. [59] use spatial neighborhood relationships and spectral similarity to efficiently identify reliable extended samples. For capturing local–global dependencies, Ghanbari et al. [60] introduced the superpixels idea to generate pseudolabels. Despite the good classification outcomes achieved by the aforementioned methods, semi-supervised learning has not been sufficiently studied in the cross-domain HSIC task. In particular, semi-supervised techniques are used to address the challenges posed by limited samples in transfer learning and the classification imbalance inherent in FSL.

III. Proposed Method

The Gprior-FSN is shown in Fig. 2 which included G-B sample enhancement mechanism and G-W feature fusion

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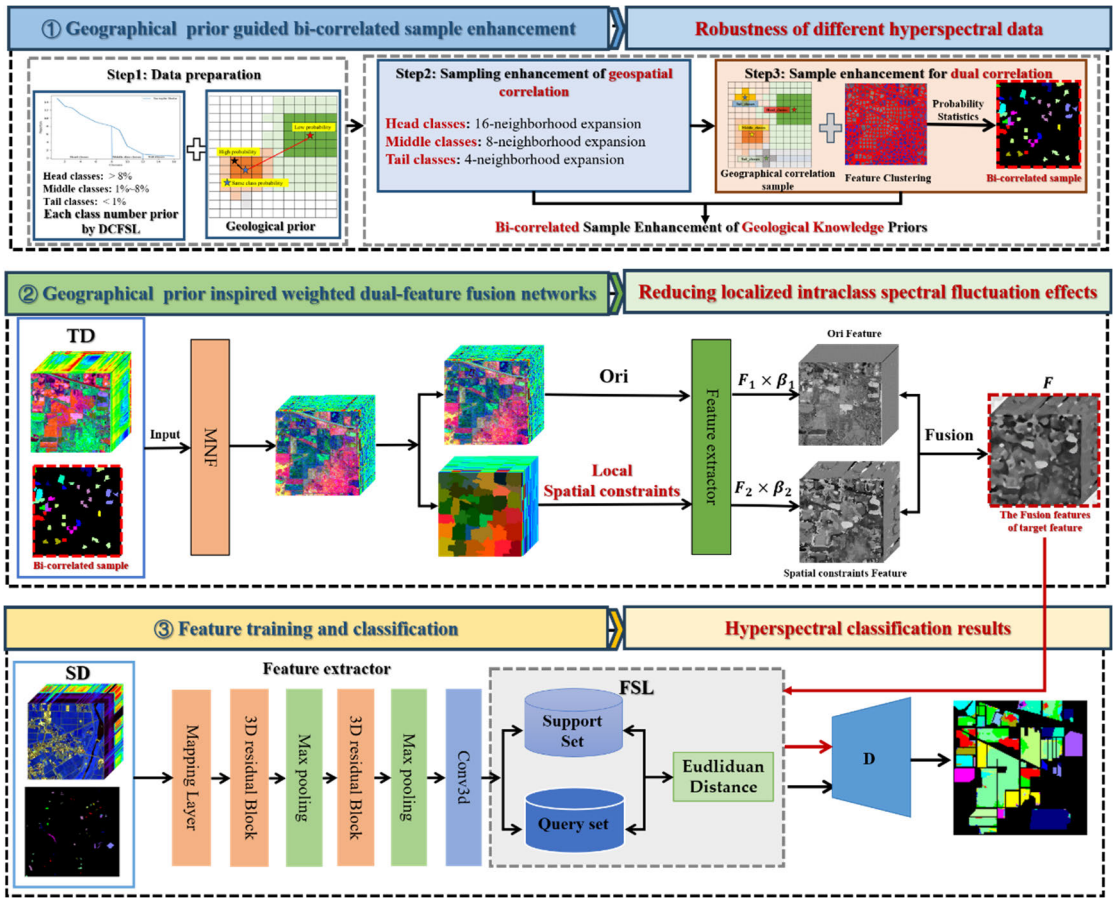


Fig. 2. Flowchart of the Gprior-FSN framework. D stands for CDD. Gprior-FSN through G-B sample enhancement mechanism to get enough high-quality pseudosamples and balance the number of per class. Gprior-FSN applies G-W dual-feature fusion network to reduce the impact of spatial local heterogeneity.

network. This method integrates semi-supervised learning, FSL, and transfer learning. The G-B sample enhancement mechanism is introduced to obtain bicorrelated pseudosamples and improve the number of head classes. G-B sample enhancement mechanism can effectively solve sample scarcity and classification imbalance of TD. MNF is used to reduce the dimensionality of TD HSI. The SD and TD HSI are simultaneously passing the mapping layer to obtain the same dimensions. The G-W feature fusion network was designed to reduce local spectral fluctuations and capture more comprehensive features of TD. CDD is applied to reduce domain shift.

A. G-B Sample Enhancement Mechanism

To fully leverage the constrained labeled data to solve sample imbalance caused by FSL method and sample insufficiency problems, the G-B sample enhancement mechanism is proposed. It aims to fully use geospatial correlation and spectral feature correlation to obtain high-confidence pseudolabels. Based on the ideas of hierarchical sample expansion and neighborhood sample expansion, geospatially correlated enhancement (GCE) is proposed. Specifically, the TD samples are first tested using the deep cross-domain FSL (DCFSL) [27] method, where the method can be replaced with any method, to obtain preliminary classification results, and then the number of samples in each category is counted. Considering that the sample imbalance in the UP and Salinas datasets is less severe compared with the IP dataset, we divided the IP dataset into three categories and the UP and Salinas datasets into two categories, based on established ranking thresholds. As shown in Fig. 1 and Table I, based on the proportion of categories to the total, categories with a percentage of more than 8% are categorized as head categories, categories with a percentage between 1.5% and 8% are categorized as middle categories, and categories with a percentage of less than 1% are categorized as tail categories. As shown in Table I, the sample imbalances in the UP and Salinas dataset are not as significant as those in the IP dataset, with the samples mainly falling into the middle and head categories. According to the first law of geography, neighboring classes have a higher likelihood of belonging to the same category. As illustrated in Fig. 2, the head classes are more numerous and are usually distributed on large parcels, hence the 16-neighborhood expansion. The middle classes are intermediate in number and its parcel size is also between the head and tail, so eight-neighborhood expansion is used. The tail classes usually have fewer parcels, so four-neighborhood expansion is used. To achieve spectral feature correlated pseudolabels, superpixel clustering algorithms and probabilistic statistical ideas are introduced. As depicted in Fig. 3, the image is initially segmented using the SLIC superpixel segmentation

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TABLE I
DCFSL METHODS PRIORI INFORMATION ABOUT THE NUMBER OF EACH CLASS

Dataset	IP			UP			Salinas		
Class	Predict	Percentage of sample number (%)	Hierarchy	Predict	Percentage of sample number (%)	Hierarchy	Predict	Percentage of sample number (%)	Hierarchy
1	66	0.64	Tail	4391	10.27	Head	2019	3.73	Middle
2	1036	10.11	Head	11630	27.19	Head	3735	6.903	Middle
3	706	6.89	Middle	2854	6.67	Middle	1973	3.643	Middle
4	682	6.65	Middle	2683	6.27	Middle	1536	2.84	Middle
5	419	4.09	Middle	1360	3.18	Middle	2535	4.68	Middle
6	750	7.32	Middle	11035	25.80	Head	4012	7.41	Middle
7	102	1.00	Tail	3382	7.91	Middle	3560	6.58	Middle
8	463	4.52	Middle	4471	10.45	Head	5879	10.86	Head
9	77	0.75	Tail	970	2.27	Middle	6225	11.5	Head
10	932	9.09	Head	967	9.54	Head	3423	6.32	Middle
11	2287	22.31	Head	---	---	---	1319	2.44	Middle
12	761	7.43	Middle	---	---	---	1956	3.61	Middle
13	224	2.19	Middle	---	---	---	943	1.7	Middle
14	1073	10.47	Head	---	---	---	1221	2.26	Middle
15	553	5.40	Middle	---	---	---	11979	22.13	Head
16	118	1.15	Tail	---	---	---	1814	3.35	Middle

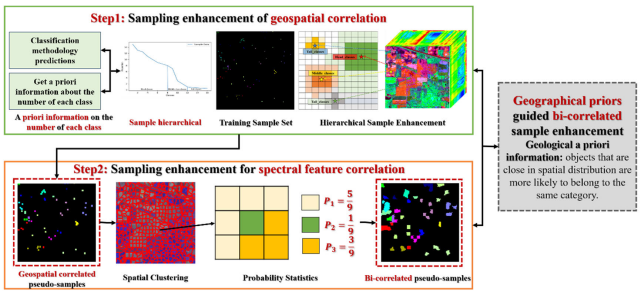


Fig. 3. G-B sample enhancement mechanism. Step 1: through the ideas of hierarchical sample expansion and neighborhood sample expansion to obtain geospatial correlated pseudolabels. Step 2: apply superpixel clustering algorithms and probabilistic statistical ideas to get spectral correlated pseudolabels.

algorithm. Subsequently, the count of each category within every superpixel block is computed through spatial similarity expansion, and each superpixel block is labeled with the category exhibiting the highest frequency. The resulting output comprises the obtained double correlated pseudolabels.

B. G-W Dual Feature Fusion Network

The incorporation of the mapping layer is designed to standardize the input sample dimensions before traversing the feature extraction network [61]. Therefore, before feature extraction, the two mapping layers M^S and M^T ensure the SD (e.g., 128 bands) and the TD (204 bands) have the same dimensions. A 2-D CNN was used for the execution of the mapping layer, and a 3-D CNN-based deep residual network was employed to extract features [62]. The detailed network structure is shown on the lower left side of Fig. 1, including two 3-D residual blocks and two maximal pooling layers, which finally pass through a 3-D convolution. In the proposed G-W feature fusion network, to reduce the computational complexity and minimize the information redundancy, the MNF method is first adopted to reduce the dimensionality of the HSI. Although HSI contains abundant spectral information, they are prone to localized spectral feature fluctuations due to noise and variations in lighting conditions. Motivated by localized spatial heterogeneity of the second law of geography, the superpixel-based spatial constraints are applied to mitigate the influence of local spectral feature fluctuations. Specifically, we use the superpixel segmentation algorithm to partition HSI features into multiple superpixel blocks. This algorithm iteratively segments the image to produce a set of spatially homogeneous superpixel regions, within which the spectral features exhibit relative uniformity. For each superpixel block, the average spectral value is computed and assigned to all the features within that block. This step creates uniform superpixel regions with similar spectral features, reducing noise from local fluctuations and improving the stability and reliability of the features. Subsequently, to comprehensively perceive the spectral features, local spatially constrained superpixel-acquired features and original features are weighted fused. This approach not only preserves the detailed information of the original spectral features but also integrates the spatial consistency provided by the superpixel blocks.

C. Few-Shot Learning and CDD

In the proposed Gprior-FSN, FSL is implemented in SD and TD simultaneously. For the SD, C classes are selected in a stochastic manner from C SD classes to constitute a set, which is a task. The support set $S_s = \{(x_n^s, y_n^s)\}_{n=1}^{C\times K}$ is constituted by selecting K samples from the designated class, while the query set $Q_s = \{(x_n^s, y_n^s)\}_{n=1}^{C\times N}$ is formed by selecting N samples from the same class. Here, S_s acts as the labeled training set, enabling the model to undergo training to minimize prediction loss on Q_S, and iterations are carried out continuously until convergence

$$L_{\text{distance}}^{S} = E\left(\left(\frac{\exp\left(-D\left(F\left(x_n^s\right), F\left(x_n^Q\right)\right)\right)}{\sum_{k=1}^{c}\exp\left(-D\left(F\left(x_n^s\right), F\left(x_n^Q\right)\right)\right)}\right), y_n^Q\right) \tag{2}$$

where F is the feature extractor, $D(,)$ is the Euclidean distance function, x_n^S is the support set sample, x_n^Q is the query set sample, and E denotes the loss of cross-entropy. For full utilization of the limited number of labeled samples, a loss function that combines the classifier prediction results and the truth labels is designed to optimize the network. The integration of these two losses with the prototype network loss together guides the training process of the model. This enhanced loss calculation method simultaneously optimizes

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the classification performance for both the support and query sets while ensuring the alignment of query set samples with the correct class prototypes

$$L^{S}_{p_r_\text{labeled}} = E\left(p\left(F\left(x_n^S\right)\right), y_n^S\right) + E\left(p\left(F\left(x_n^Q\right)\right), y_n^Q\right). \quad (3)$$

Here, p is the classifier, y_n^S is the S_S label, and y_n^Q is the Q_s label. A dual loss function update mechanism that combined the distance metric calculation results loss function and the classifier prediction label results loss function is proposed to update the entire network. Thus, the overall loss of FSL in the SD is

$$L^{s}_{\text{total_fsl}} = L^{s}_{\text{distance}} + L^{s}_{p_r_\text{labeled}}. \quad (4)$$

For TD, the same operation applies, with training samples augmented by adding random Gaussian noise to the known samples due to the limited labels. Similarly, the target overall loss of FSL in the TD is

$$L^{t}_{\text{total_fsl}} = L^{t}_{\text{distance}} + L^{t}_{p_r_\text{labeled}}. \quad (5)$$

Since most of the HSI classification task is multiclass classification, the deep network naturally has multimodal structures. Multicollinear conditioning captures cross-covariance between feature representation and classifier prediction, entropy conditioning controls uncertainty in classifier prediction to ensure moveability. Therefore, multilinear conditioning and entropy conditioning are used to capture multimodal data structures. However, multilinear conditioning is likely to cause dimensional explosion. Inspired by CDAN [63], random linear conditioning is used here instead of linear conditioning to solve the dimensional explosion. Exploring a domain discriminator D on the SD and TD distributions to mitigate domain shift. So, the loss for learning domain-invariant features is formulated as

$$\min_{D}\max_{f,g} L = -E_{x_i^s\sim p_s(x)}\log\left[D\left(f_i^s, g_i^s\right)\right] - E_{x_j^t\sim p_t(x)}\log\left[1 - D\left(f_j^t, g_j^t\right)\right] \quad (6)$$

where f represents the feature extractor and g represents the classifier. $D(,)$ is the probability of identifying as an SD sample, $1 - D(,)$ denotes the likelihood of identifying $p_t(x)$ as a TD sample, $p_s(x)$ is the SD distribution, and $p_t(x)$ is the TD distribution.

IV. Experimental Result and Analysis

To validate the effectiveness of Gprior-FSN, this article is carried out on three different TD datasets. The three TD datasets have different spectral and spatial resolutions, as well as more significant differences in classes, including agricultural and urban scene data. To prove the generalizability of Gprior-FSN, this article adopts the lychee dataset in Guangzhou area for the generalization experiments.

A. Experimental Datasets

1) SD Datasets With Large Scale and Multicategory (Chikusei Dataset): These data had 19 classes. The Headwall Hyperspec-Vnir-C sensor was applied to capture this dataset. These data consist of 2517 × 2335 pixels with 128 bands extending 343–1018 nm and a spatial resolution of 2.5 m. Sample sizes of training and testing data are shown in Table II. Fig. 4 shows the false-color (FC) image and ground-truth (GT) map.

TABLE II
Chikusei Dataset Categories and Sample Numbers

Class	Name	Total	Train
1	Water	2845	200
2	Bare soil(school)	2859	200
3	Bare soil(park)	286	200
4	Bare soil(farmland)	48525	200
5	Natural plants	4297	200
6	Weeds in farmland	1108	200
7	Forest	20516	200
8	Grass	6515	200
9	Rice field(grown)	13369	200
10	Rice field(first stage)	1268	200
11	Row crops	5961	200
12	Plastic house	2193	200
13	Manmade(non-dark)	1220	200
14	Manmade(dark)	7664	200
15	Manmade(blue)	431	200
16	Manmade(red)	222	200
17	Manmade grass	1040	200
18	Asphalt	801	200
19	Paved ground	145	--
	Total	77592	3600

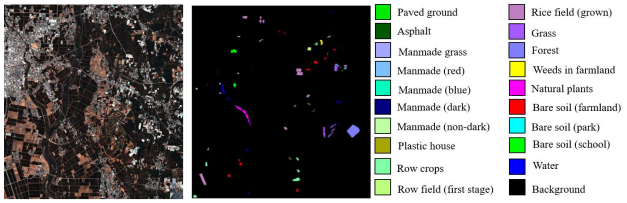


Fig. 4. Chikusei (a) FC image and (b) GT.

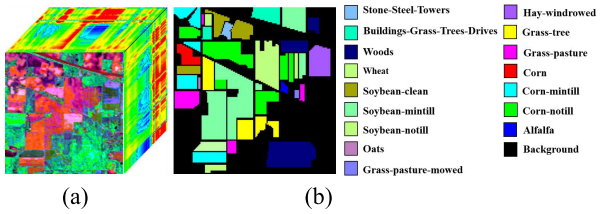


Fig. 5. IP (a) FC image and (b) GT.

2) Rural Dataset With Lower Resolution and Class Imbalanced (IPs Dataset): These data are agricultural area data with 16 classes. Received at Northwestern Indiana, AVIRS in 1992, these data consist of 145 × 145 pixels with 200 bands covering 0.4–2.5 μm and a spatial resolution of 20 m. Sample sizes of training and test data are shown in Table III. Fig. 5 shows the FC image and GT map of the IP data.

3) City Dataset With High Resolution and Samples Imbalance (UP Dataset): These data are city dataset with nine classes. The data were obtained by ROSIS sensor over the University of Pavia in 2001. The data consist of 610 × 340 pixels, 103 bands ranging from 0.43 to 0.86 μm, and a spatial resolution of 1.3 m. Table III lists the sample sizes of the training and test data. Fig. 6 shows the FC image and GT map of the UP data.

4) Rural Dataset With High Resolution and Samples Imbalance (Salinas Dataset): These data are city dataset with

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TABLE III
IP, UP, AND SALINAS DATASETS CATEGORIES AND SAMPLE NUMBERS

IP				UP			Salinas		
Class	Name	Test	Train	Name	Test	Train	Name	Test	Train
1	Alfalfa	41	5	Asphalt	6626	5	Brocoil_green_weeds_1	2004	5
2	Corn-notill	1423	5	Meadows	18644	5	Brocoil_green_weeds_2	3721	5
3	Corn-mintill	825	5	Gravel	2094	5	Fallow	1971	5
4	Corn	232	5	Trees	3059	5	Fallow_rough_plow	1389	5
5	Grass-pasture	478	5	Sheets	1340	5	Fallow_smooth	2673	5
6	Grass-tree	725	5	Bare soil	5024	5	Stubble	3954	5
7	Grass-pasture-mowed	23	5	Bitumen	1325	5	Celery	3574	5
8	Hay-windrowed	473	5	Bricks	3677	5	Grapes_untrained	11266	5
9	Oats	15	5	Shadow	942	5	Soil_vinyard_develop	6198	5
10	Soybean-notil	967	5	---	---	---	Corn_senesced_green_weeds	3273	5
11	Soybean-mintill	2450	5	---	---	---	Lettuce_romaine_4wk	1063	5
12	Soybean-clean	588	5	---	---	---	Lettuce_romaine_5wk	1922	5
13	Wheat	200	5	---	---	---	Lettuce_romaine_6wk	911	5
14	Woods	1260	5	---	---	---	Lettuce_romaine_7wk	1065	5
15	Buildings-Grass-Trees-Drives	381	5	---	---	---	Vinyard_untrained	7263	5
16	Stone-Steel-Towers	88	5	---	---	---	Vinyard_vertical_trellis	1802	5
	Total	10169	80	Total	42731	45	Total	54049	80

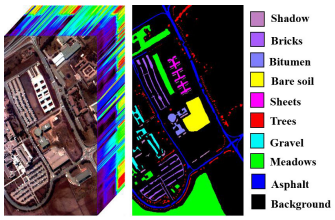


Fig. 6. UP (a) FC image (b) GT.

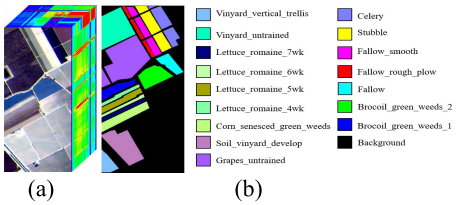


Fig. 7. Salinas (a) FC image and (b) GT.

16 classes. The ROSIS sensor is applied to acquire these data. This dataset had 512 × 217 pixels, recorded 204 spectral bands spanning 400–2500 nm. The spatial resolution measures 3.7 m. Table III shows the sample sizes of the training and test data. Fig. 7 shows the FC image and GT map.

B. Classification Performance of Gprior-FSN

All the experiments were performed on one a GeForce RTX 2080 Ti. In all, 200 labels are used for the SD, five labels are applied to the TD, and all the experiments are performed ten times using different randomized seed experiments. To adequately train the model, the experiment was performed for 20 000 iterations. At each training, the query set is set to 19 samples per class, while the support set is set to one sample per class. The methodology adopts the Adam optimizer, incorporating Xavier normalization for the initialization of both the convolutional and linear layers. The performance of CF-Trans is evaluated using overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). To fully validate the robustness of the methods, eight state-of-the-art methods are compared, including SVMs [8], contextual deep convolution neural network (CDCNN) [64], deep FSL (DFSL) [62], graph information aggregation cross-domain FSL (Gia-CFSL) [45], DCFSL [27], fast dense spectral–spatial convolution (FDSSC) [65], fine-grained predicates learning (FPGL) [66], and double-branch dual-attention mechanism network (DBDA) [67]). To evaluate the accuracy of Gprior-FSN, OA, AA, and Kappa were used.

1) Experiment 1 (IP Dataset): The quantitative classification results for the IP dataset are presented in Table IV. From Table IV, it can be seen that our method improves OA by 4.67%, AA by 17.66%, and Kappa by 5.13% on the IP dataset compared DBDA method. The IP dataset is a very fine-grained crop dataset with rich feature classes, where the problem of local spectral fluctuations is particularly important. As seen through the classification results, our method presents smoother classification results, which further proves the importance of G-W feature fusion network. Looking at the covariance matrix in Fig. 8, it is clear that the main misclassification occurs between corn and soybean because the features of corn and soybean are very similar. As seen in Table IV, Gprior-FSN shows significant enhancement in the classification of the two categories, especially achieving the best results in the Head-3 (Corn-mintill), Middle-4 (Corn), and Middle-12 (Soybean-clean) categories. For example, the classification accuracy of Corn-mintill is elevated by around 12% in contrast to other methods which further proves the effectiveness of the G-B sample enhancement mechanism. The experimental results demonstrate that Gprior-FSN outperforms current state-of-the-art cross-domain methods, including SCFormer, Gia-CFSL, DCFSL, and ADAFSL, in classification accuracy. This indicates that issues such as local spectral volatility, sample scarcity in the TD, and classification imbalance in FSL methods continue to constrain the advancement of cross-domain hyperspectral classification.

This dataset belongs to an agricultural area dataset with low resolution and extremely unbalanced samples. As shown in Fig. 9(b)–(l), the classification results using SVM, CDCNN, DFSL, SCFmorer, Gia-CFSL, DCFSL, FDSSC, FPGL, ADAFSL, DBDA, and Gprior-FSN are shown, respectively. It can also be seen from the visualization that adding spatial information methods is smoother than the classification result

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TABLE IV
QUANTITATIVE OUTCOMES (%) OF CLASSIFICATION FOR THE IP DATASETS

Class	SVM	CDCNN	DFSL	SCFormer	Gia-CFSL	DCFSL	FPGL	ADAFSL	FDSSC	DBDA	Gprior-FSN
1	19.44	21.98	47.16	85.61	89.51	90.73	92.68	98.05	74.23	60.28	**100.00**
2	38.89	32.99	48.06	45.31	40.98	43.99	50.95	53.83	65.53	**69.94**	67.44
3	25.75	25.84	46.84	52.56	52.56	54.21	54.18	60.56	52.79	55.22	**71.77**
4	17.33	20.32	30.95	62.11	78.49	76.59	74.35	82.24	50.63	51.8	**95.26**
5	48.65	33.69	82.70	71.59	71.42	76.61	73.74	78.45	**90.89**	89.12	82.22
6	83.24	53.40	91.61	90.40	84.18	85.21	87.45	84.92	**97.67**	96.45	88.62
7	21.36	20.61	36.52	96.96	99.13	98.26	**100.00**	99.13	34.40	27.20	**100.00**
8	89.91	80.17	99.41	87.02	85.67	84.12	**100.00**	87.36	99.85	99.82	96.98
9	11.77	12.91	31.82	98.67	99.33	100	**100.00**	100.00	21.87	16.39	**100.00**
10	35.10	23.65	45.91	49.79	61.75	62.81	**72.08**	69.72	65.06	68.1	71.69
11	54.23	43.07	64.98	60.05	63.34	61.09	60.27	61.48	78.44	**79.63**	63.19
12	20.33	19.92	35.21	35.94	42.93	47.14	50.68	57.84	56.70	53.87	**75.29**
13	77.72	65.92	75.93	99.55	99.30	97.15	**99.50**	95.75	93.63	89.05	94.75
14	82.41	68.69	93.19	90.31	76.52	82.68	88.21	86.66	96.54	**97.01**	87.17
15	35.11	34.63	52.77	64.54	69.24	74.28	**90.94**	74.72	67.05	72.34	86.46
16	84.50	50.51	87.44	98.52	95.45	99.89	**100.00**	98.18	73.52	68.44	96.36
OA	44.96±4.15	36.56±9.64	59.08±3.74	64.29±3.14	64.33±3.15	65.85±2.82	69.79±0.06	69.95±2.98	70.23±3.47	71.37±4.30	**76.04±2.73**
AA	46.61±2.97	38.02±8.55	60.66±2.02	74.31±2.90	75.61±2.11	77.17±1.94	80.94±0.01	80.56±1.55	69.92±4.04	68.42±3.11	**86.08±1.77**
Kappa	38.65±4.28	29.87±9.26	54.29±3.85	59.44±3.64	59.77±3.31	61.52±3.03	65.73±0.01	66.20±3.16	66.68±3.62	67.95±4.48	**73.08±2.99**

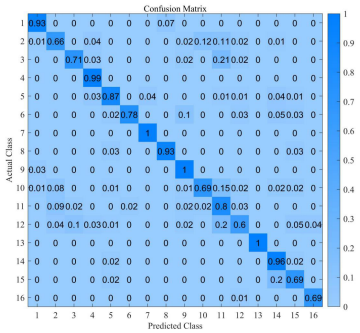


Fig. 8. Covariance matrix of IP data classification results.

of SVM. This indicates the crucial role played by spatial information. FPGL outperforms CDCNN, DFSL, Gia-CFSL, DFSL, SCFmorer, and FDSSC in classification. This is mainly because FPGL fully uses the global spatial context to capture information. The classification results unmistakably reveal that Gprior-FSN obtains significant classification improvement compared with the transfer learning methods ADAFSL, Gia-CFSL, and DCFSL. This further validates the importance of reducing local spectral fluctuation in cross-domain HSIC. According to the classification outcome, it is evident that the main error occurs between corn and soybean. This is because the spectral features of corn and soybean are very similar and the restricted quantity of training labels, and the model capturing the feature variation ability is limited. From the red box in Fig. 9, it clear that our method minimizes misclassification between corn and soybean. This further confirms the effectiveness of G-B sample enhancement mechanism. As seen in Fig. 9, our method effectively solves the classification imbalance problem on datasets with extremely unbalanced samples. From the experimental results, the classification effect of the fifth and sixth classes still has a gap compared with the optimal method FDSSC, which indicates that the convolutional dense connection of FDSSC can effectively extract the minor differences between the classes.

2) Experiment 2 (UP Dataset): The numerical classification capabilities of the UP dataset are listed in Table V. Looking at the data in Table V, it evident that Gprior-FSN framework outperforms the DCFSL method on the UP dataset, increasing OA by 3.41%, AA by 5.47%, and Kappa by 4.6%. The SD dataset Chikusei and the TD dataset UP dataset are significantly different in terms of feature classes and data distribution. They also demonstrate the validity of CDD in capturing invariant features in different domains. From the covariance matrix in Fig. 10, it can be observed that the main misclassification occurs between the categories Middle-3 (Bricks) and Head-8 (Gravel). However, Gprior-FSN achieves the best classification results in both the categories, which further validates that our method is effective in enriching the labels of the head and middle categories to effectively capture feature variations. From the experimental results, the fourth category of UP data showed a significant drop compared with other cross-domain methods. This phenomenon may be due to the more fragmented distribution of trees, which may have been mistakenly expanded during the expansion process through the bicorrelation samples, thus affecting the classification results.

The classification results of SVM, CDCNN, DFSL, SCFmorer, Gia-CFSL, DCFSL, FDSSC, FPGL, ADAFSL, DBDA, and Gprior-FSN are visualized in Fig. 11(b)–(l). The classification results unmistakably reveal that Gprior-FSN compared to the transfer learning methods SCFmorer, ADAFSL, Gia-CFSL, and DCFSL. This further validates the importance of reducing local spectral fluctuation. From the classification results, the main misclassification occurs in the header category Meadows. Following data analysis, we found that the main problem was in the Meadows area, which is sparsely vegetated with a large amount of bare ground. In addition, Meadows is a head category with a limited number of samples, which makes it difficult to capture feature variation effectively. Gprior-FSN achieves significant improvements in both the head category and the middle category, such as Meadows and Tree. It further demonstrates the effectiveness of the G-W. As shown in Fig. 11, considering the overall classification effect, our method can effectively improve the classification performance of the head class.

3) Experiment 3 (Salinas Dataset): Salinas is an agricultural area dataset with very similar feature classification categories. In addition, Salinas and the SD dataset are significantly different in terms of feature classes and data

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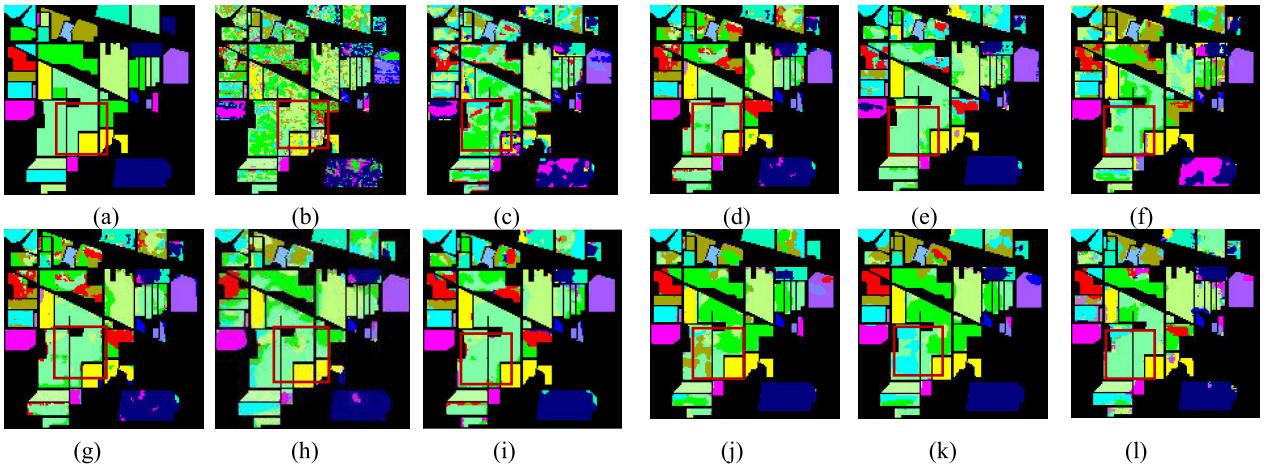


Fig. 9. Classification visualization results of IP dataset. (a) GT. (b) SVM (44.96 ± 4.15%). (c) CDCNN (36.56 ± 9.64%). (d) DFSL (59.08 ± 3.74%). (e) SCFmorer (64.29 ± 3.14%). (f) Gia-CFSL (64.33 ± 3.15%). (g) DCFSL (65.85 ± 2.82%). (h) FPGL (69.79 ± 0.06%). (i) ADAFSL (69.95 ± 2.98%). (j) FDSSC (70.23 ± 3.47%). (k) DBDA (71.37 ± 4.30%). (l) Gprior-FSN (76.04 ± 2.73%).

TABLE V
QUANTITATIVE OUTCOMES (%) OF CLASSIFICATION FOR THE UP DATASETS

Class	SVM	CDCNN	DFSL	FPGL	SCFormer	DBDA	ADAFSL	Gia-CFSL	FDSSC	DCFSL	Gprior-FSN
1	90.79	84.97	85.25	78.31	72.35	96.71	85.00	79.73	**98.41**	78.73	82.34
2	78.73	82.45	90.63	66.28	91.09	94.56	81.66	85.55	**96.01**	88.88	86.63
3	34.22	34.17	48.96	58.99	65.08	70.27	70.78	59.95	55.85	60.81	**84.95**
4	55.24	55.82	82.26	92.55	85.34	80.42	87.30	91.29	81.62	**94.94**	75.80
5	90.26	85.48	**100.00**	99.14	98.59	98.4	97.03	98.93	99.46	99.36	99.68
6	31.28	36.87	37.57	95.52	49.75	58.25	75.87	76.73	72.01	74.63	**95.52**
7	37.95	28.78	48.00	81.77	77.76	72.13	86.65	78.63	62.90	80.91	**98.25**
8	65.82	55.76	53.98	73.72	70.26	75.52	76.36	70.61	74.36	67.26	**81.54**
9	99.77	44.97	84.26	**99.84**	96.27	94.00	97.93	97.18	95.06	98.59	88.63
OA	58.70±6.22	53.60±13.89	64.6±8.05	75.99±0.06	79.78±1.66	80.92±5.86	81.91±3.78	81.94±2.65	82.62±5.05	83.12±2.38	**86.53±2.35**
AA	64.89±2.83	56.59±6.93	70.09±3.91	82.90±0.01	78.50±2.62	82.58±4.03	84.29±2.26	82.07±0.94	81.74±3.93	82.68±1.40	**88.15±1.49**
Kappa	49.18±6.42	45.10±12.86	56.70±8.83	70.09±0.06	73.12±2.12	75.87±6.93	76.65±4.33	76.62±3.14	77.84±6.12	78.01±2.71	**82.61±2.81**

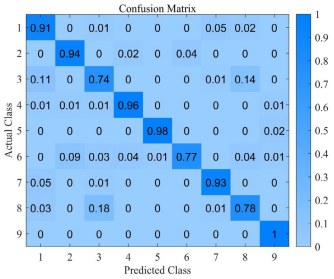


Fig. 10. Covariance matrix of UP data classification results.

distribution, with no common category. Table VI reveals that Gprior-FSN achieves the best classification accuracy, OA is 92.29%, AA is 95.13%, and Kappa is 91.42%. From the covariance matrix in Fig. 12, it can be observed that the main misclassification occurs between Head-8 (Grapes_untrained) and Head-15 (Vinyard_untrained). From analyzing the data, it was found that the features were very similar between categories 8 and 15, with a weak ability to capture feature variations due to limited label data. The Gprior-FSN framework showed excellent accuracy in these two classification tasks, improving by 2.87%–6.71% compared with the state-of-the-art cross-domain methods. This shows that the framework can effectively handle issues such as lack of samples and local spectral changes in the TD. This also demonstrates that the G-B sample enhancement mechanism is effective in solving sample scarcity and classification imbalance problems of TD.

As shown in Fig. 13(b)–(l), the classification results using CDCNN, SVM, FGPA, DFSL, SCFmorer, Gia-CFSL, FDSSC, DBDA, DCFSL, ADAFSL, and Gprior-FSN are shown, respectively. Salinas is an agricultural dataset that differs significantly from the SD dataset, with no common classes and substantial disparities in data distribution. From Gprior-FSN framework classification results, it is clear that Gprior-FSN framework visual classification results outperform other methods in terms of overall performance. The visualization results show significant improvements in the header categories "Vinyard_untrained" and "Grapes_untrained," further validating the effectiveness of G-B sample enhancement mechanism. The G-W feature fusion network has successfully achieved good classification on the Salinas dataset, which fully validates its effectiveness.

C. Discussion of Gprior-FSN Performance

1) Experiments With Different Numbers of Labeled Samples: To assess the superior performance of the Gprior-FSN method across varying numbers of training samples, a comparison of classification accuracies under various sample numbers is performed. For fairness of the experiments, all the experiments were executed on GeForce RTX 2080 Ti and with the same randomized seed. In this experiment, for each category we train using 1-5 labeled samples separately, and then compared the OA, AA, and Kappa coefficient. As illus-

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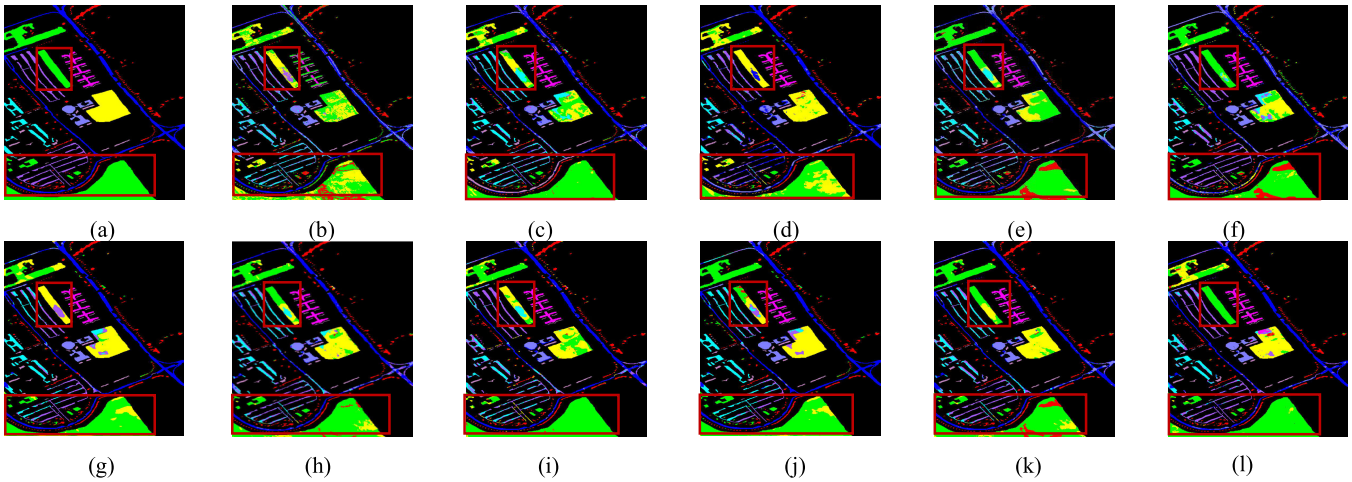


Fig. 11. Classification visualization results of the UP dataset. (a) GT. (b) SVM (58.70 ± 6.22%). (c) CDCNN (53.60 ± 13.89%). (d) DFSL (64.6 ± 8.05%). (e) FPGL (75.99 ± 0.06%). (f) SCFormer (79.78 ± 1.66%). (g) DBDA (80.92 ± 5.86%). (h) ADAFSL (81.91 ± 3.78%). (i) Gia-CFSL (81.94 ± 2.65%). (j) FDSSC (82.62 ± 5.05%). (k) DCFSL (83.12 ± 2.38%). (l) Gprior-FSN (86.53 ± 2.81%).

TABLE VI
QUANTITATIVE OUTCOMES (%) OF CLASSIFICATION FOR THE SALINAS DATASETS

Class	CDCNN	SVM	FPGL	DFSL	SCFormer	Gia-CFSL	FDSSC	DBDA	DCFSL	ADAFSL	Gprior-FSN
1	26.75	97.87	**100.00**	99.92	97.97	98.62	94.77	99.73	99.15	99.12	98.57
2	37.89	98.75	**100.00**	99.61	98.57	98.75	99.93	99.22	99.72	99.54	99.61
3	57.02	81.52	**100.00**	91.66	89.85	88.57	97.33	96.25	90.50	95.07	95.32
4	79.13	97.31	99.10	98.36	99.44	99.32	94.08	92.92	99.50	**98.96**	**99.49**
5	86.23	95.55	93.42	92.74	95.27	89.49	95.52	**96.91**	93.16	95.00	95.52
6	96.31	99.63	98.36	**99.99**	98.96	98.68	99.93	99.62	98.95	99.07	99.67
7	80.43	94.51	99.97	**99.92**	98.98	98.47	99.66	97.91	99.22	93.69	99.74
8	49.06	68.83	27.00	81.62	66.44	71.72	85.42	82.53	74.50	**73.59**	**85.85**
9	81.55	98.19	**100.00**	97.54	99.61	98.09	98.94	98.7	99.43	99.22	99.11
10	68.00	79.67	68.62	91.82	86.51	84.54	93.35	**93.94**	85.18	86.56	87.31
11	30.80	75.51	**100.00**	62.30	87.76	95.24	88.82	93.31	97.86	98.31	99.92
12	68.12	86.06	99.17	98.41	99.12	98.89	98.67	99.44	**99.73**	99.39	98.52
13	78.17	84.52	**100.00**	99.81	99.22	98.54	99.11	99.37	99.31	98.83	95.66
14	56.53	77.51	**99.62**	99.58	97.79	96.90	91.65	92.57	98.67	98.46	92.69
15	27.44	51.63	**87.82**	55.28	61.88	71.80	61.83	65.98	77.20	79.44	77.61
16	58.74	80.02	93.54	96.18	94.78	90.39	98.65	**99.22**	89.61	90.85	97.56
OA	57.35±10.43	80.29±2.16	80.52±0.03	84.40±3.22	85.58±2.16	87.33±3.07	88.06±2.71	88.94±2.43	89.35±2.16	**89.42±3.07**	**92.29±1.07**
AA	61.39±9.50	85.44±1.54	91.66±0.03	91.55±0.68	92.01±1.21	92.38±1.77	93.60±1.82	94.23±0.87	93.86±1.1	**94.07±1.70**	**95.13±1.29**
Kappa	53.49±10.71	78.18±2.32	78.57±0.04	82.77±3.47	83.98±2.40	85.94±3.37	86.76±3.0	87.71±2.65	89.01±2.38	**88.25±3.38**	**91.42±1.19**

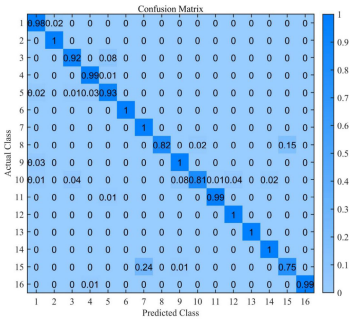


Fig. 12. Covariance matrix of Salinas data classification results.

trated in Fig. 14(a)–(c), Gprior-FSN demonstrates a substantial enhancement in classification performance for head classes compared with middle and tail classes. From the results, the OA, AA, and Kappa of the head class are significantly improved on the IP, UP, and Salinas datasets. According to Fig. 15(a)–(i), it is observed that the Gprior-FSN framework exhibits a gradual improvement as the count of labeled labels increases.

2) Ablation Experiments: Since there are significant differences between the Salinas dataset and the SD without any shared categories, we chose the Salinas dataset to measure the performance impact of individual modules within our approach. The IP dataset is selected as the second subject for ablation experiments owing to its low resolution and extremely imbalanced samples. To maintain experimental fairness, all the experiments were completed under the same experimental setup parameters. As shown in Table VII, ablation experiments were conducted on the key components of Gprior-FSN. The introduction of the D-W dual-feature fusion network led to enhancements in OA, AA, and Kappa, with improvements ranging from 1.07% to 8.62%, 1.06% to 7.9%, and 0.3% to 9.71%, respectively. The experimental results show that the D-W dual-feature fusion network is used to reduce the local spectral fluctuation in the TD, which can significantly improve the intraclass consistency and interclass discrimination. By introducing the G-B sample enhancement mechanism, the maximum improvement in OA, AA, and Kappa is 9.02%, 3.74%, and 9.71%, respectively, by incorporating the G-B sample enhancement mechanism.

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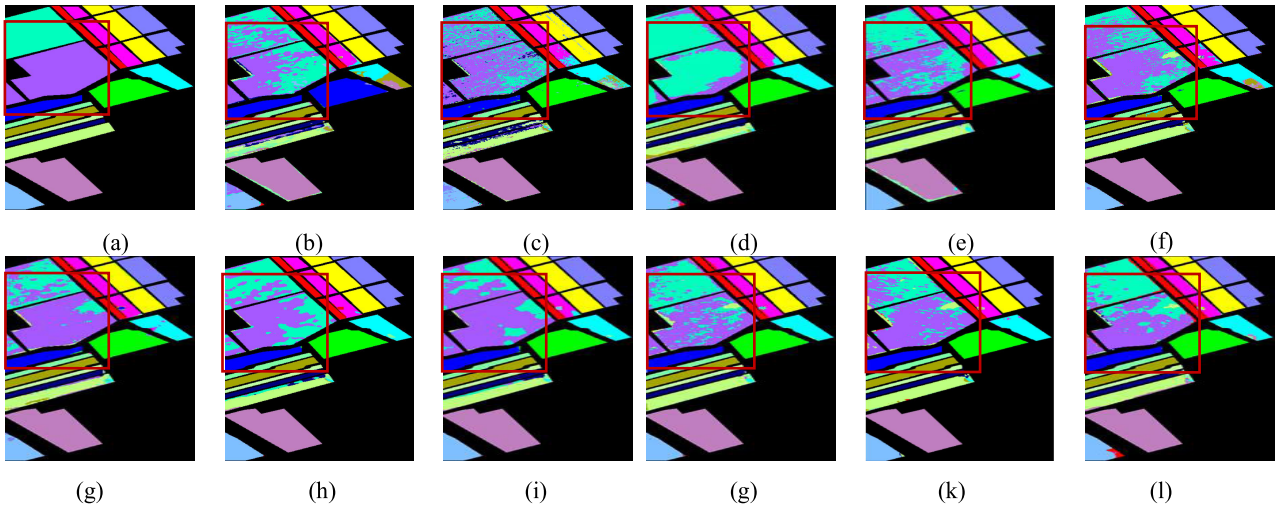


Fig. 13. Classification visualization results of the Salinas dataset. (a) GT. (b) CDCNN (57.35 ± 10.43%). (c) SVM (80.29 ± 2.16%). (d) FPGL (80.52 ± 0.03%). (e) DFSL (84.40 ± 3.22%). (f) SCFormer (85.58 ± 2.16%). (g) Gia-CFSL (87.33 ± 3.07%). (h) FDSSC (88.06 ± 2.71%). (i) DBDA (88.94 ± 2.43%). (g) DCFSL (89.35 ± 2.16%). (k) ADAFSL (89.42 ± 3.07%). (l) Gprior-FSN (92.29 ± 1.07%).

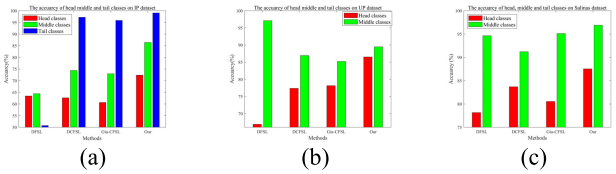


Fig. 14. Accuracy of head, middle, and tail classes on the IP, UP, and Salinas datasets. Our method significantly improves the classification of the tail classes while maintaining the classification accuracy of the head classes and middle classes. (a) Accuracy of head, middle and tail classes on IP. (b) Accuracy of head and middle classes on UP. (c) Accuracy of head and middle classes on Salinas.

TABLE VII

CLASSIFICATION OUTCOMES FROM THE ABLATION EXPERIMENT ON THE IP, UP, AND SALINAS DATASETS

Dataset	Methods	Backbone	Backbone + G-W	Backbone + G-W + G-B
IP	OA (%)	59.09	67.71	76.73
	AA (%)	75.21	83.11	86.85
	Kappa (%)	54.53	64.24	73.88
Salinas	OA (%)	89.35	90.40	93.09
	AA (%)	93.86	92.69	94.33
	Kappa (%)	89.01	89.32	92.31

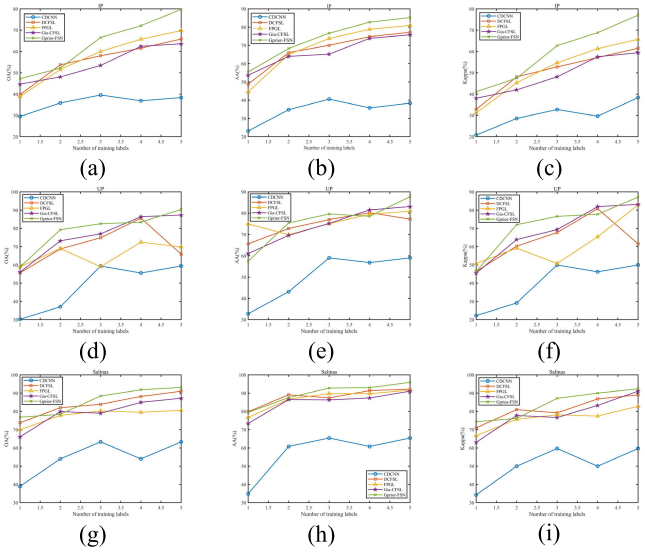


Fig. 15. Classifying with varied labeled sample sizes. The Gprior-FSN framework demonstrates incremental enhancement with the escalation of labeled instances. Comparison outcomes against alternative methodologies consistently demonstrate our approach achieves superior classification results in the majority of cases; (a)–(c), (d)–(f), and (g)–(l) show the OA, AA, and Kappa coefficients for the IP, UP, and Salinas datasets, respectively, under varying labeled sample sizes.

3) Feature Aggregation Performance: To enhance the interpretability of Gprior-FSN, we have visualized feature maps. The Gia-CFSL method was used to analyze feature aggregation, which combines graph convolution and FSL. As observed in Fig. 16(a), (d), and (g), the IP, UP, and Salinas datasets display intricate confusion in their original feature distributions. Fig. 16(b), (e), and (h) clearly shows the most significant confusion between category 10 (Soybean-notil) and category 12 (Soybean-clean) in the IP dataset, 1 (Asphalt) and 5 (Sheets) in the UP dataset, 7 (Celery), and 14 (Lettuce_romaine_7wk) in the Salinas dataset. It can be clearly observed through Fig. 16(c), (f), and (l) that the confusion between these categories is significantly reduced. This provides further evidence of the enhanced performance of Gprior-FSN in feature extraction.

4) Computing Efficiency of Gprior-FSN: To validate the efficiency of Gprior-FSN, the training, testing time, FLOPs, and parameters of the Gprior-FSN were compared with four cross-domain HSIC methods. All the experiments were performed on GeForce RTX 2080 Ti to ensure the fairness of our experiments, and all the experiments apply same GPU. As shown in Table VIII, the Gprior-FSN has significantly reduced training and testing time. Compared with other methods, Gprior-FSNexhibits a significant reduction in both FLOPs and the number of parameters. On the IP dataset, the training time of Gprior-FSN is approximately 0.5 times that of ADAFSL, 0.18 times that of Gia-CFSL, 0.31 times that of HFSL, and 0.58 times that of DCFSL. Concurrently, Gprior-FSN demonstrates a significant reduction than other methods in both FLOPs and parameter number. On the UP

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TABLE VIII
TRAINING AND TESTING TIME ANALYSIS

Datasets	Computing time	DCFSL	HFSL	Gia-CFSL	ADAFSL	Gprior-FSN
IP	Train time (s)	2833.64	5327.66	9025.32	3127.98	**1638.52**
	Test time (s)	1.88	4.14	1.79	1.79	**1.3**
	FLOPs (G)	0.664158720	3.210143424	0.666567168	3.727616512	**0.015734160**
	#params (M)	0.07	3.30	0.22	**0.02**	0.03
UP	Train time (s)	1620.32	2737.20	3527.15	2149.48	**1027.69**
	Test time (s)	5.92	25.54	7.19	6.87	**3.46**
	FLOPs (G)	0.664158720	3.196163216	0.666567168	3.727616512	**0.083915520**
	#params (M)	0.07	3.16	0.22	0.02	**0.03**
Salinas	Train time (s)	2510.56	6450.31	5843.49	12736.74	**1518.52**
	Test time (s)	8.27	64.83	9.76	23.51	**4.63**
	FLOPs (G)	0.66415870	3.210723456	0.666567168	3.727616512	**0.083915520**
	#params (M)	0.07	3.31	0.22	0.02	**0.03**

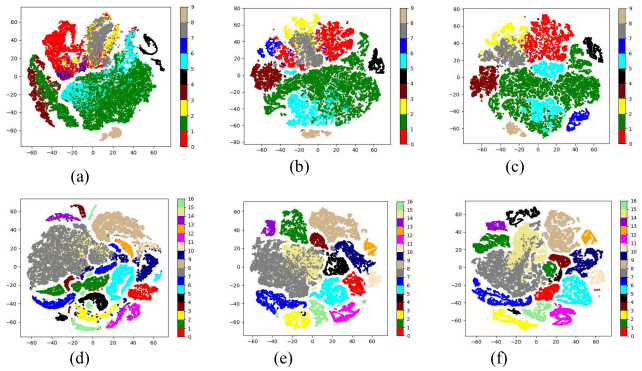


Fig. 16. Feature aggregation performance of the IP, UP, and Salinas datasets. (a) UP original features. (b) Gia-CFSL extraction features. (c) Gprior-FSN extraction features. (d) Salinas original features. (e) Gia-CFSL extraction features. (f) Gprior-FSN extraction features.

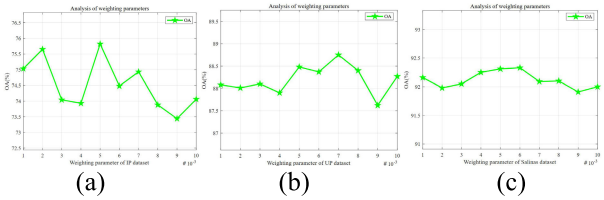


Fig. 17. Weighting parameter analysis. (a) Classification result with various weights for local spatial constraints on the IP dataset. (b) Classification result with various weights for local spatial constraints on the UP dataset. (c) Classification result with various weights for local spatial constraints on the Salinas dataset.

dataset, the training time of Gprior-FSN is approximately 0.5 times that of ADAFSL, 0.29 times that of Gia-CFSL, 0.38 times that of HFSL, and 0.63 times that of DCFSL. In addition, Gprior-FSN exhibits a significant reduction in FLOPs compared with the state-of-the-art ADAFSL method, with FLOPs being approximately 0.023 of those of ADAFSL. On the Salinas dataset, Gprior-FSN requires only 26% of the training time needed by Gia-CFSL, 24% of the time needed by HFSL, and 60% of the time needed by DCFSL. The training and testing times demonstrate a substantial enhancement in the efficiency of Gprior-FSN.

5) Analysis of Weighting Parameters: During the experimental process, the weight parameters of the local spatial constraint features are finely tuned and optimized several times in this article. By systematically adjusting the weighting parameter and analyzing the experimental results, we found that setting the weighting parameter to 0.005 leads to more stable performance. As shown in Fig. 17, a weight of 0.005 yielded the best classification results on the IP dataset, demonstrating its superior performance in handling complex scenes. On the UP dataset, optimal classification results were achieved with weighting parameters of 0.005, 0.006, and 0.007. On the Salinas dataset, classification performance was optimal with weight parameters of 0.005 and 0.006. Based on these analyses, a weighting parameter of 0.005 was ultimately selected for the experiment.

D. Generalization Experiment of Gprior-FSN

To verify the generalization ability of Gprior-FSN, we conducted experiments on a GF5 hyperspectral dataset located in Guangzhou. To facilitate comparison to other methods, we incorporated a comparative analysis of the DBDA and Gia-CFSL approaches into the generalization experiment. This dataset has 150 bands with a spatial resolution of 30 m and image range of 380 × 375 pixels. The resolution of this dataset is low, it is difficult to distinguish features, and the distribution of lychees is irregular, which is extremely challenging. Therefore, this dataset was selected to be more representative for generalization verification of the method. In these Fig. 18, the black area represents the background, while the green color indicates the litchi. The experimental setup for this experiment is the SD using the Chikusei dataset with 200 labels per category. TD uses only five labeled lychees image samples. The distribution of lychees in the dataset is scattered, and it is very challenging to obtain labels from low-resolution hyperspectral satellite images. Therefore, when using the G-B sample enhancement mechanism, we set smaller superpixel parameters and only used four-neighborhood expansion. The experimental results show that the overall classification results are relatively complete despite some errors in Gprior-FSN. As shown in Fig. 18, Gprior-FSN markedly reduces the false-negative rate in classifying lychees with fragmented distributions compared with the DBDA and Gia-CFSL methods. Although these data mainly consist of irregular, scattered features, the classification results show that the method performs well in identifying these lychees, demonstrating its strong generalization ability. Especially the small plots of lychees in the red box can also be identified, demonstrating the powerful engineering application capabilities of Gprior-FSN. The G-B sample enhancement mechanism proposed in this article performs well on category-rich and sample-unbalanced datasets. However, with appropriate adjustments, the mechanism can also achieve good results in practical engineering applications.

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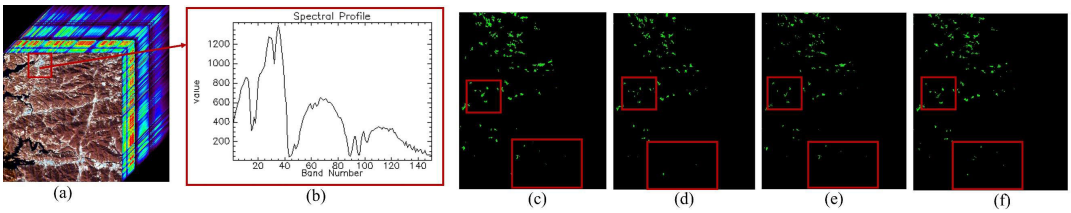


Fig. 18. Generalization performance of Gprior-FSN method. (a) GF-5 HSI with 150 bands and 30-m spatial resolution. (b) Spectral curves of litchi. (c) GT. (d) Classification map of DBDA. (e) Classification map of Gia-CFSL. (f) Classification map of Gprior-FSN.

V. Conclusion

This article proposed a Gprior-FSN for cross-domain HSIC. This framework mainly consists of G-B sample enhancement mechanism and G-W feature fusion network. The G-B sample enhancement mechanism is proposed to solve sample scarcity and FSL classification imbalance problem. To make the acquired pseudolabels geospatially correlated, incorporating the first law of geography, which suggests that spatially proximate categories tend to belong to the same class, this study introduces neighborhood sample expansion to enhance the sample size. Furthermore, considering the category imbalance of using FSL-based method, a hierarchical sample idea was applied to solve the sample imbalance problem. To correlate the acquired pseudolabels in spectral features, the superpixel clustering method is introduced. For the acquired pseudolabels to be bicorrelated, pseudosamples were counted for each category within each superpixel block. Each superpixel block is assigned the feature category with the highest frequency. To minimize the effects of local intraclass spectral fluctuations, G-W feature fusion network is proposed. The G-W feature fusion network uses local spatial constraints to mitigate intraclass variance, thereby enhancing interclass discrimination. To validate Gprior-FSN method reliability, Gprior-FSN was compared with the state-of-the-art methods on three datasets including agricultural and urban area data. To validate the generalizability of the method, we performed generalization experiments on the real GF5 HSI data. The experimental results indicate that Gprior-FSN framework can well solve sample problem and local intraclass spectral fluctuations problem. FSL represents a highly effective approach for addressing the issue of unseen classes in TD. However, challenges stemming from sample scarcity and classification imbalance continue to constrain its classification performance. Moving forward, we aim to undertake further exploration to overcome these challenges.

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